

FURTHER RESULTS ON THE NONLINEAR SEPARATION PRINCIPLE: THE GENERAL 'ASYMPTOTICALLY CONTROLLABLE' CASE

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Abstract: It is shown, for general nonlinear systems, that asymptotic controllability and observability are sufficient for semiglobal practical asymptotic stabilization by output feedback. Indeed, as previously shown in the literature, asymptotic controllability implies the existence of a (discontinuous in general) state feedback that, when implemented by sample and hold, is semiglobally practically stabilizing and robust to measurement disturbance; moreover, a weak form of observability allows reconstruction of the state with arbitrary precision in an arbitrarily short amount of time. So, we can build an output feedback that operates periodically in two modes: an initial, small fraction of a sampling period is used to estimate the state, and the remainder of the sampling period is used to implement the state feedback control using the state estimate. Our stabilization results are presented not only for a compact target set (e.g. the origin) but also for general closed and nonempty target sets. *Copyright ©2001 IFAC*

Keywords: Asymptotic controllability, Output feedback, Separation principle

1. INTRODUCTION

It is well-known that semiglobal stabilization by output feedback can be achieved for nonlinear systems that are (globally) asymptotically stabilizable by *smooth* state feedback and are (globally) *uniformly* observable (Esfandiari and Khalil, 1992; Khalil and Esfandiari, 1993; Teel and Praly, 1994; Atassi and Khalil, 1999; Shim and Seo, 2000; Maggiore and Passino, 2000). Global stabilization by output feedback is not possible, in general, without adding to these assumptions, as demonstrated by the counterexample of Mazenc *et al.* (1994).

In this paper we show that semiglobal (practical) stabilization by output feedback is achievable even when the assumptions above are relaxed to global asymptotic controllability and non-uniform observability. For convenience, we begin with the definition of asymptotic controllability taken from the paper of Kellett and Teel (2000). Here, we consider a system given by

$$\dot{x} = f(x, u), \quad x \in \mathbb{R}^n, u \in \mathbb{R}^m, \quad (1a)$$

$$y = h(x, u), \quad y \in \mathbb{R}^p. \quad (1b)$$

where f and h are smooth in x and u .

Definition 1. (Asymptotic Controllability). Let a set $\mathcal{A} \subset \mathbb{R}^n$ be closed and nonempty. The system

(1a) is *asymptotically controllable to the set $\mathcal{A}$* if there exist a $\mathcal{KL}$ function $\beta(\cdot, \cdot)$ and a nondecreasing function $\rho_c(\cdot)$ such that: for each $x \in \mathbb{R}^n$ there exists a measurable function $u_c : [0, \infty) \rightarrow \mathbb{R}^m$ such that the solution $x(\cdot)$ of (1a) with $u = u_c$ exists and satisfies

$$\begin{aligned} \|x(t)\|_{\mathcal{A}} &\leq \beta(\|x(0)\|_{\mathcal{A}}, t) \\ \|u_c(t)\| &\leq \rho_c(\|x(t)\|_{\mathcal{A}}) \end{aligned}$$

for all $t \geq 0$ where $\|x\|_{\mathcal{A}}$ denotes $\inf_{y \in \mathcal{A}} \|x - y\|$.

While, for linear systems, it is well-known how to construct stabilizing linear state 'feedback' based only on asymptotic controllability (i.e. stabilizability) to the origin characterized using open-loop control $u_c(\cdot)$, results on constructing state feedback for nonlinear systems are much more recent. In fact, the equivalence between asymptotic controllability and the existence of a (possibly discontinuous) state feedback stabilizer has been established by not sticking to continuous feedback (see (Clarke *et al.*, 1997; Clarke *et al.*, 2000; Kellett and Teel, 2000); also see a nice tutorial of Sontag (1999b) and references therein).

We will rely on the results of Kellett and Teel (2000) where, like in (Clarke *et al.*, 1997; Clarke *et al.*, 2000; Sontag, 1999a), a state feedback scheme has been constructed, based on asymptotic controllability, that guarantees semiglobal practical stability of the set $\mathcal{A}$

for the system (1a). The control synthesis algorithm used there constructs a control Lyapunov function (clf) first and, then a control value is selected that guarantees some decrease of the value of clf. We can denote the selection algorithm by a feedback function $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^m$. When implemented, the selected control value $\alpha(x)$ is applied to the system for a fixed time period. Therefore, the overall state feedback control strategy has the nature of sampled-data control.

However, the function $x \mapsto \alpha(x)$ is not necessarily continuous and the related clf is not necessarily differentiable, which makes building an output feedback controller based on certainty equivalence—replacing the state in α by the state's estimate—more challenging. Fortunately, the nature of sampled-data control produces some robustness to measurement disturbances (see also Sontag (1999b; 1999a) and Kellelt and Teel (2000, Sec. 5)) even under a discontinuous state feedback, and it also enables the analysis based on the difference of a Lyapunov function, which will be utilized in this paper.

For output feedback stabilization of (1) some type of observability is required. The observability proposed in this paper is the following:

Definition 2. (Observability). The system (1) is said to have *semiglobal finite-dimensional robust observability (SGFDRO)* if, for each $l_x \geq 0$, there exist a time $\bar{t}_o > 0$, an integer $q \geq 0$, a C^{q+1} function $u^*(\cdot) : [0, \bar{t}_o] \rightarrow \mathbb{R}^m$ and a function $\Psi : \mathbb{R}^{(q+1)p+(q+1)m} \rightarrow \mathbb{R}^n$ such that, for the system $\dot{x}(t) = f(x(t), u^*(t))$, $y(t) = h(x(t), u^*(t))$ with initial conditions $x(0)$ satisfying $\|x(0)\|_{\mathcal{A}} \leq l_x$, the followings hold: for $t \in [0, \bar{t}_o]$,

- the solution $x(t)$ exists,
- $x(t) = \Psi(Y(t), U^*(t))$ where Y and U^* represent $Y := [y^T, \dot{y}^T, \dots, (y^{(q)})^T]^T$ and $U^* := [(u^*)^T, (\dot{u}^*)^T, \dots, ((u^*)^{(q)})^T]^T$, respectively,
- for each $\epsilon \geq 0$, there exists $\delta \geq 0$ such that

$$\begin{aligned} \|\hat{Y} - Y(t)\| &\leq \delta \quad \Rightarrow \\ \|\Psi(\hat{Y}, U^*(t)) - \Psi(Y(t), U^*(t))\| &\leq \epsilon \quad (2) \end{aligned}$$

where $\hat{Y} \in \mathbb{R}^{(q+1)p}$.

This condition is much weaker than the well-known ‘uniform observability’ (Gauthier and Bornard, 1981). Roughly speaking, while uniform observability requires that $x = \Psi(Y, U)$ holds for *any* input $u(\cdot)$, our condition implies the reconstruction of state by the output and a *specific* input $u^*(\cdot)$. This point is in sharp contrast with the previous output feedback results (Teel and Praly, 1994; Shim and Seo, 2000), but has already been exploited by Coron (1994) for the output feedback stabilization problem in the case where continuous, possibly time-varying, state feedback stabilizers are available. Our definition of observability is stronger than the generic nonlinear

observability defined by Hermann and Krener (1977), but will turn out to permit finite dimensional, approximate observers. Discussions on our observability may be found in (Shim and Teel, 2001) which contains several illustrative examples for this observability and complete proofs of lemmas in this paper.

Our goal is to achieve semiglobal practical stabilization of the system by output feedback when the asymptotic controllability and our observability conditions hold. At this point, it is interesting to notice that, roughly stated, asymptotic controllability and observability require that $\alpha(x(t))$ and $u^*(t)$ should be applied to the system, respectively. We resolve this conflict by employing a so-called periodic ‘dual mode’ operation in our output feedback strategy; that is, a part of a sampling period is devoted to observing the state and the rest is for stabilizing the system. We establish this ‘separation principle’ when stabilizing not only the origin but also general noncompact target set $\mathcal{A}$. Therefore, the terms ‘semiglobal’ and ‘practical’ should be interpreted in the sense of the distance from the set $\mathcal{A}$. In order to make the presentation clear, in section 2 we first state our result for the case where the set $\mathcal{A}$ is compact, and then state its extension to the noncompact $\mathcal{A}$ case. In section 3, we provide the proof of our results using assumptions that generalize those made in section 2. Our conclusions can be found in section 4.

Throughout our work, $\mathbb{Z}^+$ denotes the set of non-negative integers, and $\|\cdot\|$ denotes the Euclidean norm for a vector and the induced Euclidean norm for a matrix. For a function $V : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ and constants r and R such that $0 \leq r \leq R$, we define $\mathcal{V}(r, R) := \{x \in \mathbb{R}^n : r \leq V(x) \leq R\}$. We denote $\mathcal{V}(0, R)$ by $\mathcal{V}(R)$. By abuse of notation, $L_f h(x, U^*)$ implies (x, U^*) -space Lie derivative, that is, $L_f h(x, U^*) := \frac{\partial h}{\partial x}(x, u^*)f(x, u^*) + \frac{\partial h}{\partial u^*}(x, u^*)\dot{u}^*$, and $L_f^i h$ ($i \geq 2$) is defined recursively.

2. MAIN RESULTS

The form of proposed output feedback controller, which achieves the semiglobal practical asymptotic stabilization to the set $\mathcal{A}$, is given as follows:

$$\dot{\hat{Y}}(t) = \begin{cases} A\hat{Y}(t) + \theta \Delta_\theta H(y(t) - C\hat{Y}(t)), & \text{for } iT \leq t < iT + \tilde{t} \\ 0, & \text{for } iT + \tilde{t} \leq t < (i+1)T \end{cases} \quad (3)$$

$$\chi(i) = \Psi(\hat{Y}(iT + \tilde{t}), U^*(\tilde{t})) \quad (4)$$

$$u(t) = \begin{cases} u^*(t - iT), & \text{for } iT \leq t < iT + \tilde{t} \\ \alpha(\chi(i)), & \text{for } iT + \tilde{t} \leq t < (i+1)T \end{cases} \quad (5)$$

where $i \in \mathbb{Z}^+$, $T > 0$ is a sampling time, $\tilde{t} > 0$ is such that $\tilde{t} < T$ and $\tilde{t} \leq \bar{t}_o$, $\hat{Y} \in \mathbb{R}^{(q+1)p}$ and $\chi \in \mathbb{R}^n$. Let I be the identity matrix of dimension p . Then, $C = [I, 0, \dots, 0]$, $H = [a_1 I, a_2 I, \dots, a_{q+1} I]^T$ in which a_i 's are chosen such that the polynomial

$s^{q+1} + a_1 s^q + a_2 s^{q-1} + \dots + a_{q+1}$ is Hurwitz, $\Delta_\theta = \text{diag}\{I, \theta I, \theta^2 I, \dots, \theta^q I\}$ with constant θ , and

$$A = \begin{bmatrix} 0 & I & 0 & \dots & 0 \\ 0 & 0 & I & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & I \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}. \quad (6)$$

Appropriate values of $T, \tilde{t}, \theta, q$ and functions u^* and Ψ will be specified subsequently. Note that (4) shows the sample and hold nature of our controller, and $\alpha(\cdot)$ in (5) is a feedback law for the system designed with the state of system, but possibly discontinuous with respect to the state x . The initial condition of the observer $\hat{Y}(0)$ should be put in $\Sigma(y(0))$, where $\Sigma(y)$ is the set of admissible initial conditions for the observer which possibly depends on the measured output y . $\Sigma(y)$ will also be specified subsequently with the proof that, if $\hat{Y}(iT) \in \Sigma(y(iT))$, then $\hat{Y}((i+1)T) \in \Sigma(y((i+1)T))$. Note also that the definition of observer (3) on the interval $[iT + \tilde{t}, (i+1)T]$ is somewhat arbitrary and just needs to guarantee that $\hat{Y}((i+1)T) \in \Sigma(y((i+1)T))$.

A typical control action of the proposed output feedback is depicted as solid lines in Figure 1. In every time period $[iT, iT + \tilde{t}]$ the input to the system is the fixed signal $u^*(\cdot)$ determined by the observability condition. While $u^*(\cdot)$ is applied, the observer (3) operates in order to yield sufficiently close estimate $\chi(i)$ to x at $t = iT + \tilde{t}$ with which the control value $\alpha(\chi(i))$ will be calculated. However, since this input u^* is not necessarily a part of the stabilizing action, $\tilde{t}$ will be chosen small in order for u^* not to damage the goal of stabilization. Accordingly, θ will be chosen for sufficiently fast convergence of the observer (3) (i.e. high-gain observer) during the time period $[iT, iT + \tilde{t}]$. However, while $\chi(i)$ will be close to $x(iT + \tilde{t})$, there is no reason that $\alpha(\chi(i))$ is close to $\alpha(x(iT + \tilde{t}))$ since continuity of $\alpha(\cdot)$ is not assumed. Nevertheless, we can handle this difficulty by using the continuity of f in an appropriate way. One thing we rely on is the robustness of state feedback to measurement disturbance in an $\mathcal{L}_\infty$ sense since the mismatch between $\chi(i)$ and $x(iT + \tilde{t})$ can be thought of as measurement disturbance (also see (Sontag, 1999b; Sontag, 1999a) and Kellett and Teel (2000, Sec. 5)). The other robustness needed is to the action of u^* . We will see that it can be thought of an actuator disturbance that is small, not in an $\mathcal{L}_\infty$ sense but in an $\mathcal{L}_1$ sense, and that the state feedback is robust to this type of disturbance as well.

2.1 With the compactness of $\mathcal{A}$

When the set $\mathcal{A}$ is compact, our result can be concisely stated as follows.

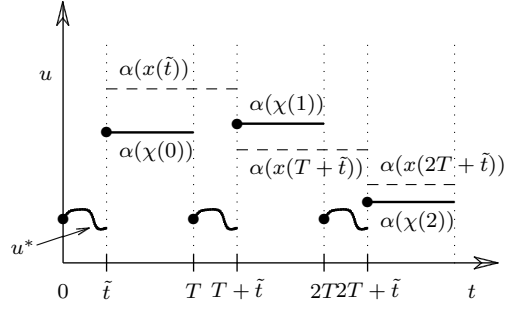


Fig. 1. A typical output feedback control action

Theorem 1. Suppose that the system (1) is asymptotically controllable to a compact set $\mathcal{A}$ and observable (SGFDRO). Then there exist a function $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^m$, a locally Lipschitz function $V : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ and class- $\mathcal{K}_\infty$ functions α_1 and α_2 such that $\alpha_1(\|x\|_{\mathcal{A}}) \leq V(x) \leq \alpha_2(\|x\|_{\mathcal{A}})$ for all $x \in \mathbb{R}^n$ and, for each pair of constants (r, R) satisfying $0 < 2r < R$, there exist a map Ψ , an input u^* with suitable degree q , a compact set Σ of admissible initial conditions for the observer (3) and positive constants $T, \tilde{t}, \theta$ and c such that the $x(\cdot)$ component of the solution to the closed-loop (1), (3)–(5) with an initial condition $(x(0), \hat{Y}(0)) \in \mathcal{V}(R) \times \Sigma$, satisfies

$$V(x(t)) \leq \max\{V(x(0)) - k(t)c, r\} + r, \text{ for } t \geq 0$$

where $k(t)$ is the largest integer k such that $kT \leq t$, and $V(x((i+1)T)) \leq V(x(iT))$ if $x(iT) \in \mathcal{V}(r, R)$ for each $i \in \mathbb{Z}^+$.

The inequality of the theorem shows that $V(x(t))$ goes below the value of $2r$ in finite time and stays there. Therefore, by taking r sufficiently small and R sufficiently large, the semiglobal practical asymptotic stabilization to the set $\mathcal{A}$ is achieved by output feedback.

2.2 Without the compactness of $\mathcal{A}$

The claims of Theorem 1 can still be derived even when the set $\mathcal{A}$ is not compact if we assume the main consequences of compactness that are used to establish Theorem 1. Firstly, the following assumption is for the construction of *state* feedback which has already been exploited by Kellett and Teel (2000) for the systems asymptotically controllable to a non-compact set $\mathcal{A}$.

Assumption 1. There exists a nondecreasing and locally Lipschitz function $\rho(\cdot)$ such that $\rho(s) \geq \rho_c(s)$ for all $s \geq 0$, and for each $l_x \geq 0$ and $l_\rho \geq 0$,

- (E1) $\exists M$ such that $\|f(x, u)\| \leq M$ for all $(x, u) \in S(l_x, l_\rho) := \{(x, u) : \|x\|_{\mathcal{A}} \leq l_x, \|u\| \leq \rho(l_\rho)\}$,
- (E2) $\exists L$ such that $\|f(x_1, u_1) - f(x_2, u_2)\| \leq L\|x_1 - x_2\| + L\|u_1 - u_2\|$ for all $(x_1, u_1) \in S(l_x, l_\rho)$ and $(x_2, u_2) \in S(l_x, l_\rho)$.

Now, we propose the following assumption for the output feedback. Obviously, both assumptions are satisfied when $\mathcal{A}$ is compact.

Assumption 2. In addition to the observability of the system (1),

(E3) for each $l_x \geq 0$, $\exists M$ s.t. $\|f(x, u^*(t))\| \leq M$ and $\|L_f h(x, u^*(t), \dot{u}^*(t))\| \leq M$,

$$\begin{aligned} & \vdots \\ & \|L_f^{q+1} h(x, u^*(t), \dots, (u^*)^{(q+1)}(t))\| \leq M, \\ & \text{for all } x \text{ such that } \|x\|_{\mathcal{A}} \leq l_x \text{ and } t \in [0, \bar{t}_o], \end{aligned}$$

(E4) for each $l_x \geq 0$ and $l_p \geq 0$, $\exists M$ such that $\|(\partial h / \partial x)(x, u)\| \leq M$ for all $(x, u) \in S(l_x, l_p)$.

If Assumptions 1 and 2 are satisfied then the conclusion of Theorem 1 holds with Σ now a set-valued map depending on $y(0)$ whose value is nonempty and compact.

Theorem 2. Suppose the system (1) is asymptotically controllable and observable (SGFDRO). If Assumptions 1 and 2 are satisfied, then there exist a function $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^m$, a locally Lipschitz function $V : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ and class- $\mathcal{K}_{\infty}$ functions α_1 and α_2 such that $\alpha_1(\|x\|_{\mathcal{A}}) \leq V(x) \leq \alpha_2(\|x\|_{\mathcal{A}})$ for all $x \in \mathbb{R}^n$ and, for each pair of constants (r, R) satisfying $0 < 2r < R$, there exist a map Ψ , an input u^* with suitable degree q , a set-valued map $y \mapsto \Sigma(y)$ such that $\Sigma(y)$ is a nonempty and compact subset of $\mathbb{R}^{p(q+1)}$, and positive constants $T, \tilde{t}, \theta$ and c such that the $x(\cdot)$ component of the solution to the closed-loop (1), (3)–(5) with an initial condition $(x(0), \hat{Y}(0)) \in \mathcal{V}(R) \times \Sigma(y(0))$, satisfies

$$V(x(t)) \leq \max\{V(x(0)) - k(t)c, r\} + r, \text{ for } t \geq 0$$

where $k(t)$ is the largest integer k such that $kT \leq t$, and $V(x((i+1)T)) \leq V(x(iT))$ if $x(iT) \in \mathcal{V}(r, R)$ for each $i \in \mathbb{Z}^+$.

Since (E1)–(E4) hold when $\mathcal{A}$ is compact, Theorem 1 follows trivially from Theorem 2 except the existence of Σ not depending on y which will be shown in Remark 1.

3. PROOF OF THEOREM 2

3.1 Robustness with State Feedback

From now on, we proceed with the following assumptions (Assumptions 3, 4 and 5) which are automatically satisfied from asymptotic controllability, observability and (E1)–(E4). In fact, while (A4) of Assumption 3 follows directly from (E2), the conditions (A1)–(A3) can be derived from the consequence of Kellett and Teel (2000) where a (possibly non-differentiable) Lyapunov function V and a (possibly discontinuous) state feedback α are constructed with sampling period T for asymptotically controllable

systems. Due to the limit of space we omit the full links to Assumption 3 but the readers might be encouraged to see Kellett and Teel (2000). (We also omit the proofs of all the lemmas.)

Assumption 3. There are a nonnegative continuous function V such that $\mathcal{V}(r, R)$ is nonempty, a state feedback control α , and positive constants T, c, L, σ and ϵ ($3\epsilon < r$) such that

(A1) $V(x(T)) \leq \max\{V(x_0) - (c + 3\epsilon), (r - 3\epsilon)\}$ if $x_0 \in \mathcal{V}(r - 3\epsilon, R + \epsilon)$

(A2) $V(x(t)) \leq \max\{V(x_0), r\} + (r - 3\epsilon)$ for $t \in [0, T]$ if $x_0 \in \mathcal{V}(R + \epsilon)$

(A3) $\|x - w\| < \sigma$ implies $|V(x) - V(w)| < \epsilon$ for $x, w \in \mathcal{V}(R + r + \epsilon)$

(A4) $\|f(x, \alpha(z)) - f(w, \alpha(z))\| \leq L\|x - w\|$ for $x, w, z \in \mathcal{V}(R + r + \epsilon)$

where $x(t)$ is the solution trajectory of $\dot{x} = f(x, \alpha(x_0))$ with $x(0) = x_0$.

The following lemma shows that, if we sacrifice a little amount of the size of target sublevel sets, the bound of inter-sample behavior and the convergence rate, then the given state feedback induces robustness with respect to additive disturbances that are small in an $\mathcal{L}_1$ sense.

Lemma 1. Suppose Assumption 3 holds for (1a). Let $D = \sigma / (T \exp(LT))$. Then the solution $x(t)$ of

$$\dot{x}(t) = f(x(t), \alpha(x(0))) + d(t), \quad x(0) \in \mathcal{V}(R + \epsilon),$$

with $d(\cdot)$ such that

$$\frac{1}{T} \int_0^{\hat{t}} \|d(s)\| ds < D, \quad \text{for some } \hat{t} \in (0, T],$$

satisfies

$$V(x(t)) \leq \max\{V(x(0)), r\} + (r - 2\epsilon),$$

for $t \in [0, \hat{t}]$. Furthermore, when $\hat{t} = T$ and $x(0) \in \mathcal{V}(r - \epsilon, R + \epsilon)$, the following inequality holds:

$$V(x(T)) \leq \max\{V(x(0)) - (c + 2\epsilon), (r - 2\epsilon)\}.$$

We append to Assumption 3 the following conditions (A5) and (A6), which obviously follow from (E1) and (E3).

Assumption 4. There exist M_{α} and M^* such that

(A5) $\|f(x, \alpha(z))\| \leq M_{\alpha}, \forall x \in \mathcal{V}(R + r), z \in \mathcal{V}(R + r + \epsilon)$,

(A6) $\|f(x, u^*(t))\| \leq M^*, \forall x \in \mathcal{V}(R + r), t \in [0, \bar{t}_o]$.

Now, with D given by Lemma 1, we choose $\tilde{t}$ ($0 < \tilde{t} \leq \bar{t}_o$ and $< T$) and $\sigma_o > 0$ such that

$$TL\sigma_o + ((1 + TL)M^* + M_{\alpha})\tilde{t} < DT \quad (7)$$

$$\sigma_o + M^*\tilde{t} < \sigma. \quad (8)$$

This choice will be justified subsequently.

The rest of proof will show the following claims: there exist a constant θ and a set-valued map Σ with nonempty, compact values, such that the solution $x(\cdot)$ of (1) with (3)–(5) satisfies that, if $\hat{Y}(iT) \in \Sigma(y(iT))$,

- (O1) $V(x((i+1)T)) \leq \max\{V(x(iT)) - c, r\}$ if $x(iT) \in \mathcal{V}(r, R)$,
- (O2) $V(x(t)) \leq \max\{V(x(iT)), r\} + r$ for $t \in [iT, (i+1)T)$ if $x(iT) \in \mathcal{V}(R)$,
- (O3) $\hat{Y}((i+1)T) \in \Sigma(y((i+1)T))$,

for all $i \in \mathbb{Z}^+$. From (O1)–(O3), the proof of Theorem 2 is completed.

3.2 Convergence of Observer

To be more specific, we use the following Assumption 5 which also follows from (E3), (E4) and the observability. Let $\bar{U}^* := [(U^*)^T, ((u^*)^{(q+1)})^T]^T$.

Assumption 5. Suppose that the observability condition holds. For given $\tilde{t}$, σ_o and ϵ ,

- (H1) $\exists$ a compact set Σ_b such that

$$[L_f h^T(x, U^*(t)), \dots, L_f^q h^T(x, U^*(t))]^T \in \Sigma_b$$

for all $x \in \mathcal{V}(R + \epsilon)$ and $t \in [0, \tilde{t}]$,

- (H2) $\exists M_y$ such that $\|(\partial h / \partial x) f(x, \alpha(z))\| \leq M_y$ for $x \in \mathcal{V}(R + r)$ and $z \in \mathcal{V}(R + r + \epsilon)$,

- (H3) $\exists M_\phi$ such that $\|\phi(x, \bar{U}^*(t))\| \leq M_\phi$ for $x \in \mathcal{V}(R + \epsilon)$ and $t \in [0, \tilde{t}]$ where $\phi(x, \bar{U}^*) := L_f^{q+1} h(x, \bar{U}^*)$,

- (H4) $\exists \delta_o$ s.t. $\|\hat{Y} - Y\| \leq \delta_o \Rightarrow \|\Psi(\hat{Y}, U^*(\tilde{t})) - \Psi(Y, U^*(\tilde{t}))\| \leq \sigma_o$ for all $Y = [h^T(x, u^*(\tilde{t})), \dots, L_f^q h^T(x, u^*(\tilde{t}))]^T$ where $x \in \mathcal{V}(R + \epsilon)$.

Taking successive derivatives of the output y of the system (1), when $u(t) = u^*(t)$, we obtain the following representation of the system:

$$\dot{Y} = AY + B\phi(x, \bar{U}^*)$$

where $B = [0, 0, \dots, 0, I]^T$ and A is defined in (6). For convenience, the observer (3) is rewritten here.

$$\begin{aligned} \dot{\hat{Y}} &= A\hat{Y} + \theta \Delta_\theta H(y - C\hat{Y}), \quad \hat{Y}(0) \in \Sigma(y(0)) \\ \hat{x} &= \Psi(\hat{Y}, U^*) \end{aligned} \quad (9)$$

Let M_Σ be a constant such that $M_\Sigma \geq \delta_o$. Now we define the set of admissible initial conditions for observer as

$$\Sigma(y) := \{Y \in \mathbb{R}^{(q+1)p} : \|Y_1 - y\| \leq M_y T + M_\Sigma, \|Y_{[2, q+1]}\|_{\Sigma_b} \leq M_\Sigma\} \quad (10)$$

where Y_1 is the first p -elements of the vector Y and $Y_{[2, q+1]}$ is the rest of the vector Y . Note that $\Sigma(y)$ for each y is compact since Σ_b is compact by (H1). The observer is initialized so that $\hat{Y}(0) \in \Sigma(y(0))$, which is possible since $y(0)$ is measurable in the beginning and Σ_b is known *a priori*.

Lemma 2. Suppose that Assumptions 3, 4 and 5 hold with $\tilde{t}$ and σ_o satisfying (7) and (8). Then, there exists $\theta \geq 1$ with which the observer (3) guarantees that, when $x(0) \in \mathcal{V}(R)$, $\hat{Y}(0) \in \Sigma(y(0))$ and $u(t) = u^*(t)$ for $t \in [0, \tilde{t}]$,

$$\|\hat{Y}(\tilde{t}) - Y(\tilde{t})\| \leq \delta_o, \quad (11)$$

which in turn implies that

$$\|\hat{x}(\tilde{t}) - x(\tilde{t})\| \leq \sigma_o. \quad (12)$$

The time-invariant structure (9) and property (12) enable the repetitive operation of the observer in conjunction with the output feedback control scheme (i.e. (3)–(5)). However, to ensure the repetitive operation of the observer, it should be shown that $x(iT) \in \mathcal{V}(R)$ and $\hat{Y}(iT) \in \Sigma(y(iT))$ for each $i \in \mathbb{Z}^+$. Here the latter is shown assuming the former which will be shown in the following subsection. Suppose that $x(iT) \in \mathcal{V}(R)$ and $\hat{Y}(iT) \in \Sigma(y(iT))$. Then, inequality (11) implies $\|\hat{Y}_1(iT + \tilde{t}) - Y_1(iT + \tilde{t})\| \leq \delta_o$ and $\|\hat{Y}_{[2, q+1]}(iT + \tilde{t}) - Y_{[2, q+1]}(iT + \tilde{t})\| \leq \delta_o$. Since $\|\dot{y}(t)\| \leq M_y$ for $t \in [iT + \tilde{t}, (i+1)T)$ by (H2) and some property for what x does over this interval, and $\hat{Y}_1((i+1)T) = \hat{Y}_1(iT + \tilde{t})$ by (3), we obtain

$$\begin{aligned} &\|\hat{Y}_1((i+1)T) - Y_1((i+1)T)\| \\ &\leq \|\hat{Y}_1(iT + \tilde{t}) - Y_1(iT + \tilde{t})\| + \|Y_1(iT + \tilde{t}) - Y_1((i+1)T)\| \\ &\leq \delta_o + M_y T. \end{aligned}$$

Similarly, $\|\hat{Y}_{[2, q+1]}(iT + \tilde{t}) - Y_{[2, q+1]}(iT + \tilde{t})\| \leq \delta_o$ implies $\|\hat{Y}_{[2, q+1]}((i+1)T)\|_{\Sigma_b} \leq \delta_o$ since $Y_{[2, q+1]}(iT + \tilde{t}) \in \Sigma_b$ and $\hat{Y}_{[2, q+1]}((i+1)T) = \hat{Y}_{[2, q+1]}(iT + \tilde{t})$. This shows the claim (O3).

Remark 1. When the set $\mathcal{A}$ is compact, we can take a compact admissible set Σ , which is independent of y . In fact, if $x(iT) \in \mathcal{V}(R)$, then, for $t \in [iT, iT + \tilde{t}]$, $x(t) \in \mathcal{V}(R + \epsilon)$ which implies that $y(t) = h(x(t), u^*(t))$ is contained in a certain compact set Σ_a from the continuity of h and the compactness of sublevel sets. Then, a compact set Σ is taken such that $\Sigma \supset \Sigma(y)$ for all $y \in \Sigma_a$. Now, for the fixed Σ , Lemma 2 again yields θ such that (11) holds. Hence, $\hat{Y}(iT) \in \Sigma$ implies $\|\hat{Y}(iT + \tilde{t}) - Y(iT + \tilde{t})\| \leq \delta_o$, which in turn leads to $\hat{Y}((i+1)T) = \hat{Y}(iT + \tilde{t}) \in \Sigma(y(iT + \tilde{t})) \subset \Sigma$. $\square$

3.3 Convergence and Boundedness of $V(x(t))$

Now it is left to show the claims (O1) and (O2). By applying the proposed output feedback controller, the state of the system evolves according to the following equations:

$$\begin{aligned} \dot{x}(t) &= f(x(t), u^*(t - iT)) \quad \text{for } t \in [iT, iT + \tilde{t}] \\ \dot{x}(t) &= f(x(t), \alpha(\chi(i))) \quad \text{for } t \in [iT + \tilde{t}, (i+1)T). \end{aligned}$$

Suppose that $x(iT) \in \mathcal{V}(R)$ and $x(t) \in \mathcal{V}(R + r)$ for $t \in [iT, (i+1)T]$. Take $\bar{\delta}_i := \chi(i) - x(iT)$. Note that, by (A6) and Lemma 2,

$$\|\bar{\delta}_i\| \leq \sigma_o + M^* \tilde{t} =: \Delta(\tilde{t}, \sigma_o), \quad (13)$$

since $\|\bar{\delta}_i\| \leq \|\chi(i) - x(iT + \tilde{t})\| + \|x(iT + \tilde{t}) - x(iT)\|$ and $\chi(i) = \hat{x}(iT + \tilde{t})$.

Let $z(t) := x(t) + \bar{\delta}_i$. Then,

$$\dot{z} = f(z, \alpha(\chi(i))) + \delta_1(t) + \delta_2(t) \quad (14)$$

where $\delta_1(t) := f(x(t), \alpha(\chi(i))) - f(x(t) + \bar{\delta}_i, \alpha(\chi(i)))$ and

$$\delta_2(t) := \begin{cases} f(x(t), u^*(t - iT)) - f(x(t), \alpha(\chi(i))), & t \in [iT, iT + \tilde{t}) \\ 0, & t \in [iT + \tilde{t}, (i+1)T). \end{cases}$$

Let $\Phi(i, t) := \int_{iT}^{iT+t} \|\delta_1(s) + \delta_2(s)\| ds$. Then, by (A3), (A4), (A5), (A6), (8) and Lemma 2, it can be shown that

$$\Phi(i, T) \leq TL\Delta(\tilde{t}, \sigma_o) + (M^* + M_\alpha)\tilde{t}. \quad (15)$$

Since σ_o and $\tilde{t}$ is chosen by (7), (15) implies that

$$\frac{1}{T} \int_{iT}^{(i+1)T} \|\delta_1 + \delta_2\| dt < D. \quad (16)$$

Therefore, by applying Lemma 1 for (14) with $\hat{t} = T$ keeping in mind that $\chi(i) = x(iT) + \bar{\delta}_i = z(iT)$, we obtain the following inequalities:

$$V(z((i+1)T)) \leq \max\{V(z(iT)) - (c + 2\epsilon), (r - 2\epsilon)\}$$

when $z(iT) \in \mathcal{V}(r - \epsilon, R + \epsilon)$, and

$$V(z(t)) \leq \max\{V(z(iT)), r\} + (r - 2\epsilon)$$

when $z(iT) \in \mathcal{V}(R + \epsilon)$ for $t \in [iT, (i+1)T]$. Thus, with (A3) and the fact $\Delta(\tilde{t}, \sigma_o) < \sigma$ by (8), we finally obtain that

$$V(x((i+1)T)) \leq \max\{V(x(iT)) - c, r\} \text{ if } x(iT) \in \mathcal{V}(r, R)$$

and

$$V(x(t)) \leq \max\{V(x(iT)), r\} + r \quad \text{if } x(iT) \in \mathcal{V}(R)$$

for $t \in [iT, (i+1)T]$, i.e., (O1) and (O2) hold.

Now we justify the temporary assumption that $x(t) \in \mathcal{V}(R + r)$ for $t \in [iT, (i+1)T]$ when $x(iT) \in \mathcal{V}(R)$. Suppose there is $t^* \in (0, T]$ such that $V(x(iT + t^*)) = R + r$ and $V(x(t)) < R + r$ for $t \in [iT, iT + t^*)$. Then, applying the similar arguments above, we get a contradiction. (Further details are omitted for the limited space.)

4. CONCLUSIONS

We have presented an output feedback control scheme which achieves semiglobal practical asymptotic stabilization for general nonlinear systems that are asymptotically controllable to a set and observable. The control input divides its time between two roles: one for the observation of the state and the other for stabilization. Our stabilization results are presented not only for a compact target set $\mathcal{A}$ but also for general closed and nonempty target sets.

This work is a generalization of the prevalent semiglobal separation principle results (Esfandiari and Khalil, 1992; Khalil and Esfandiari, 1993; Teel and Praly, 1994; Atassi and Khalil, 1999; Shim and Seo, 2000; Maggiore and Passino, 2000) in that *smooth* stabilizability and *uniform* observability have been relaxed. Also, the state feedback stabilization schemes (Sontag, 1999a; Clarke *et al.*, 2000; Kellett and Teel, 2000) for general asymptotic controllable systems have found their natural extensions to output feedback.

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